DATA VISUALIZATION FOR DECISION MAKING

PRACTICAL REPORT

Student Name: Jeel Patel

Enrollment No.:

T26084801021/26084801012

Course: M. SC.IT (Data Science)

Semester: 1

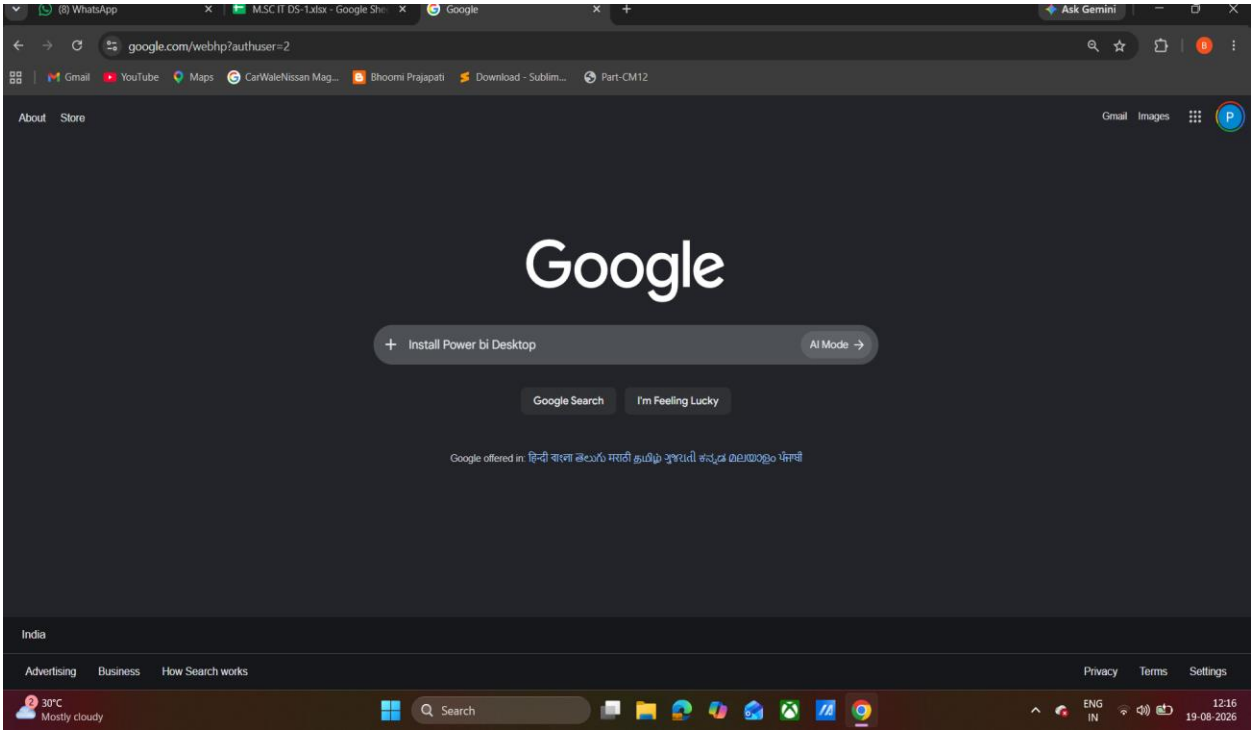
Course Code: P91A3DVD

Academic Year: 2026–27

Practical 1: Installation and Data Import

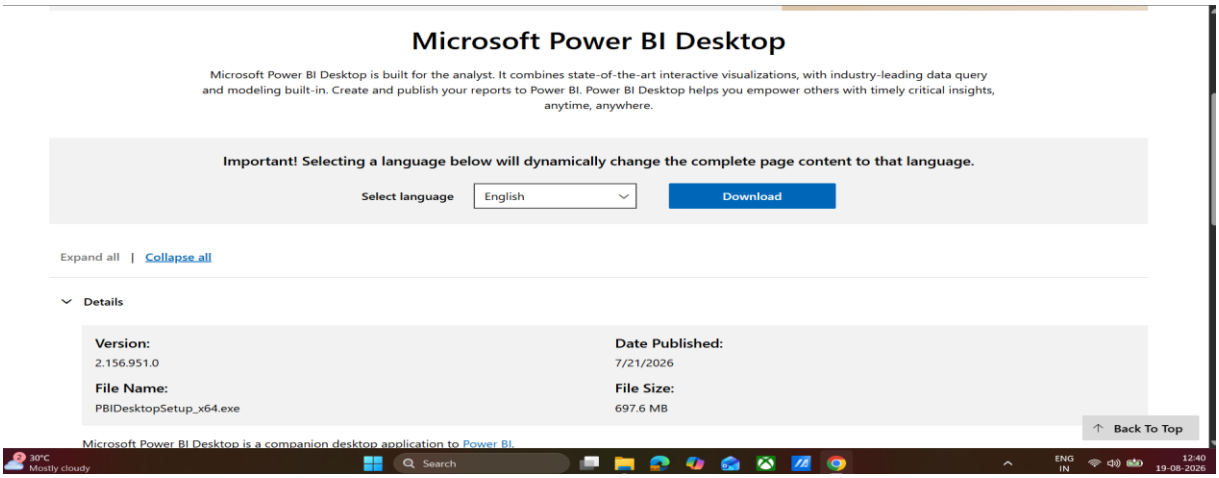
Objective: Install Power BI Desktop and import data from an Excel/CSV file.

Step 1: Download Power BI Desktop

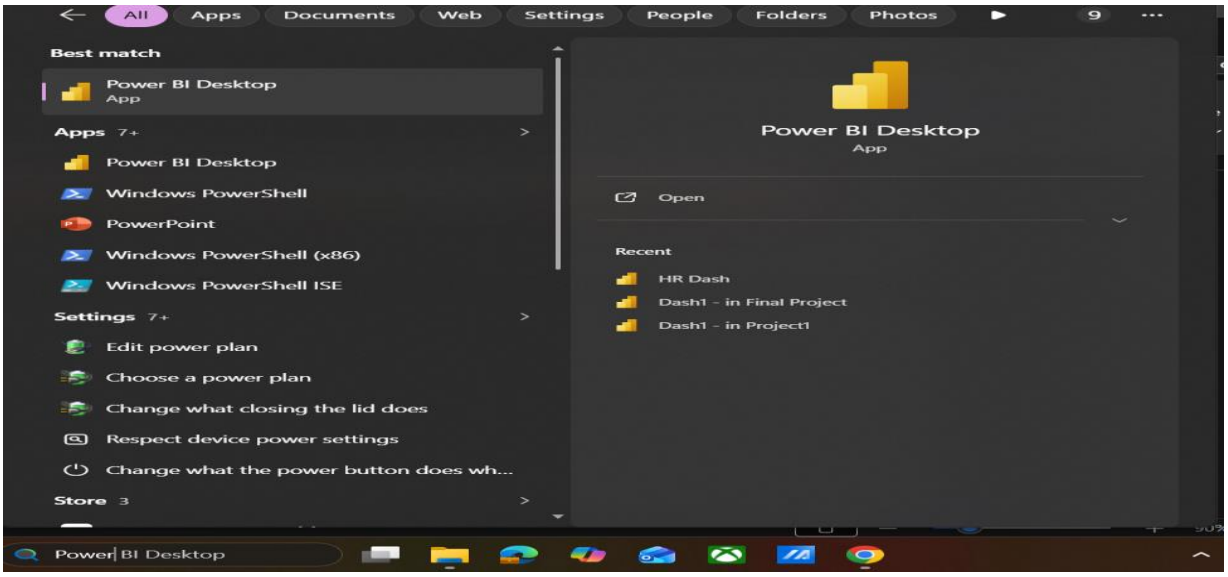
- 

- Open your web browser, search for "**install Power BI Desktop**" on Google, and navigate to the official Microsoft Power BI Desktop download page.

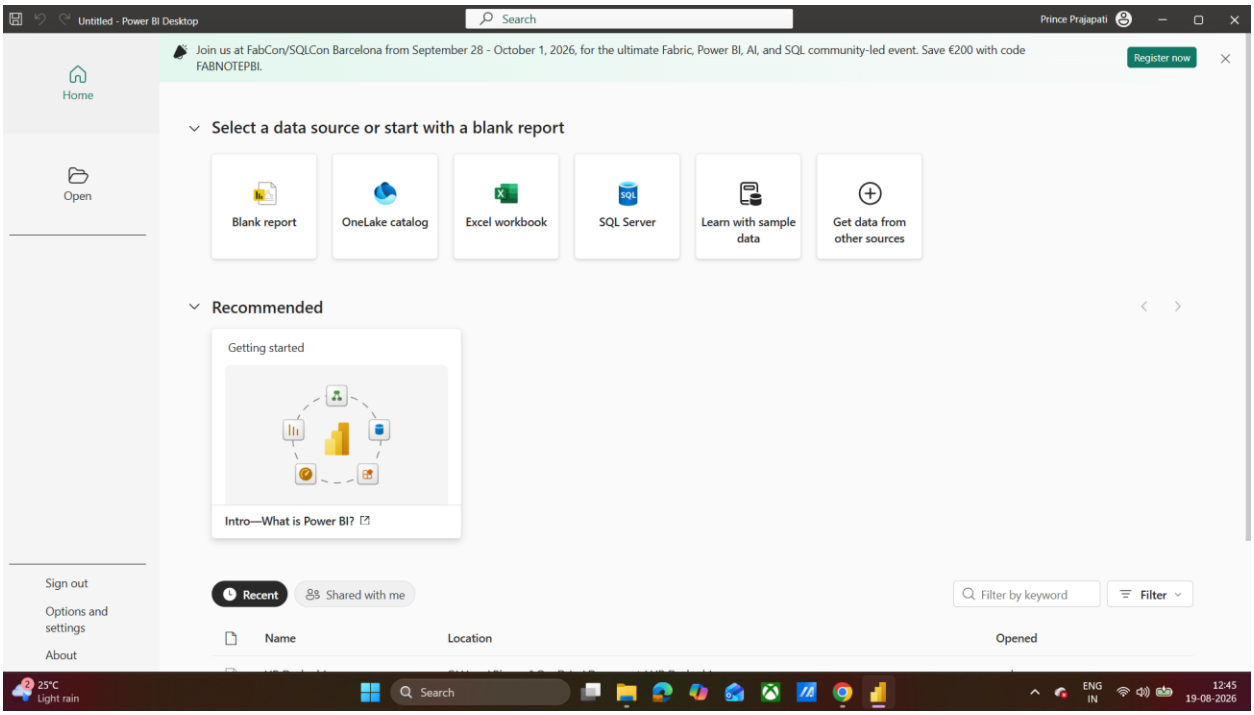
- Select English language and click Download to get the setup executable file.



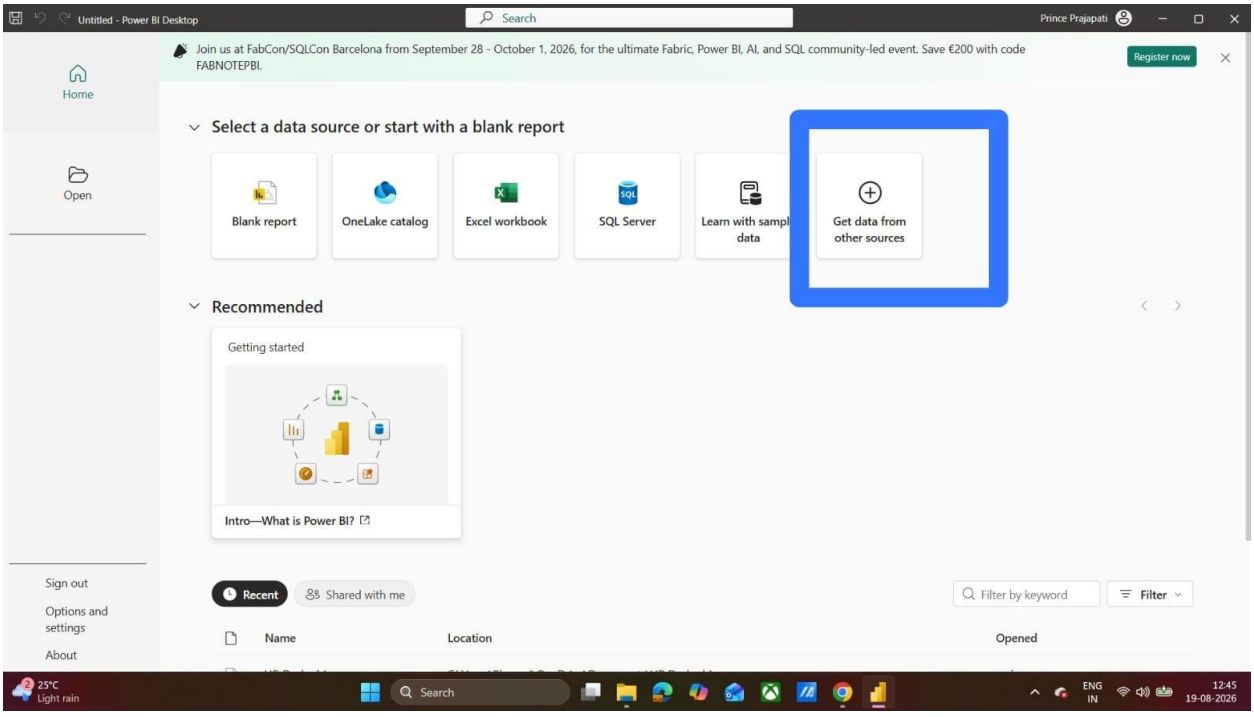
Step 2: Open Power BI Desktop



- Press the Windows Key or click the Start Menu search bar on your computer.
- Type Power BI Desktop in the search field.
- Click on the Power BI Desktop App under the search results to launch it.
- Once the application opens, close or navigate through the splash screen to view the primary welcome layout, where you can select a data source or start with a blank report.



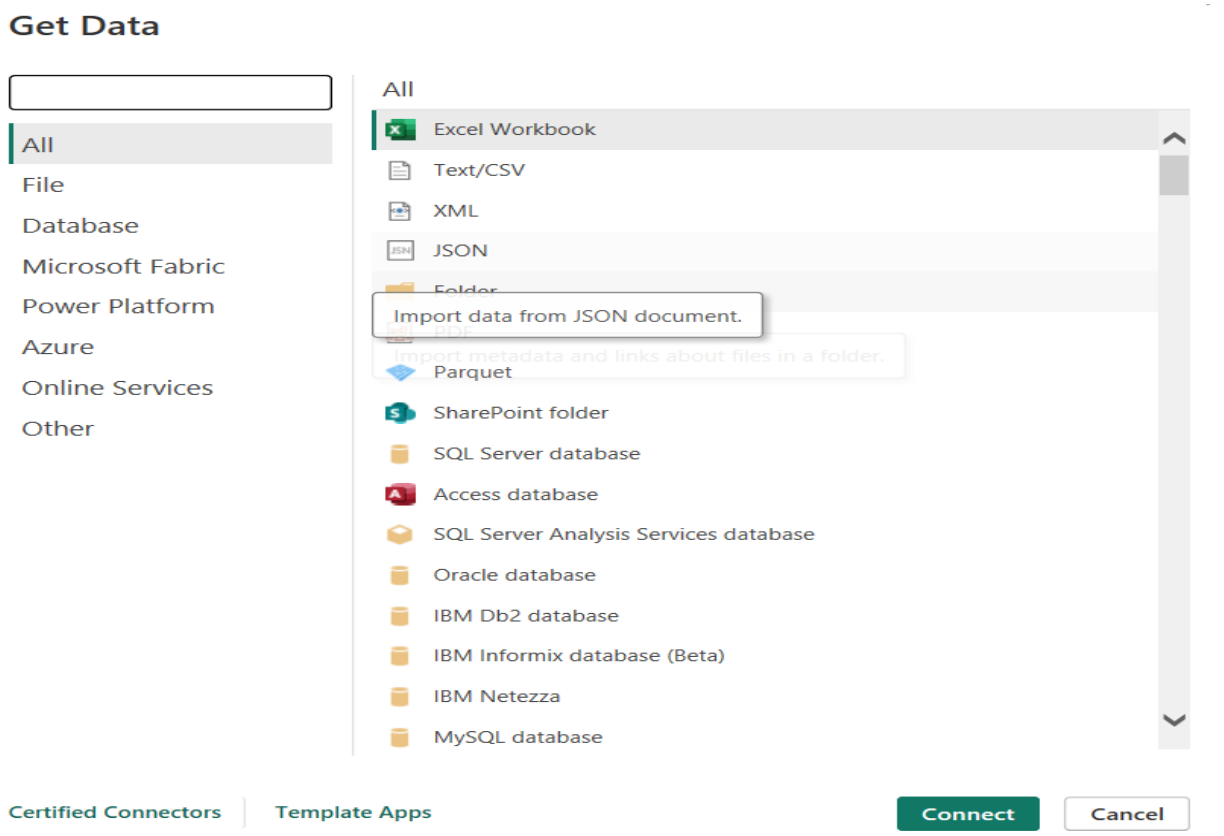
Step 3: Select Get Data



- On the Power BI Desktop welcome splash screen, locate the primary data import options under Select a data source or start with a blank report.
- Click on the Get data from other sources of tile (represented by a circle with a plus sign).

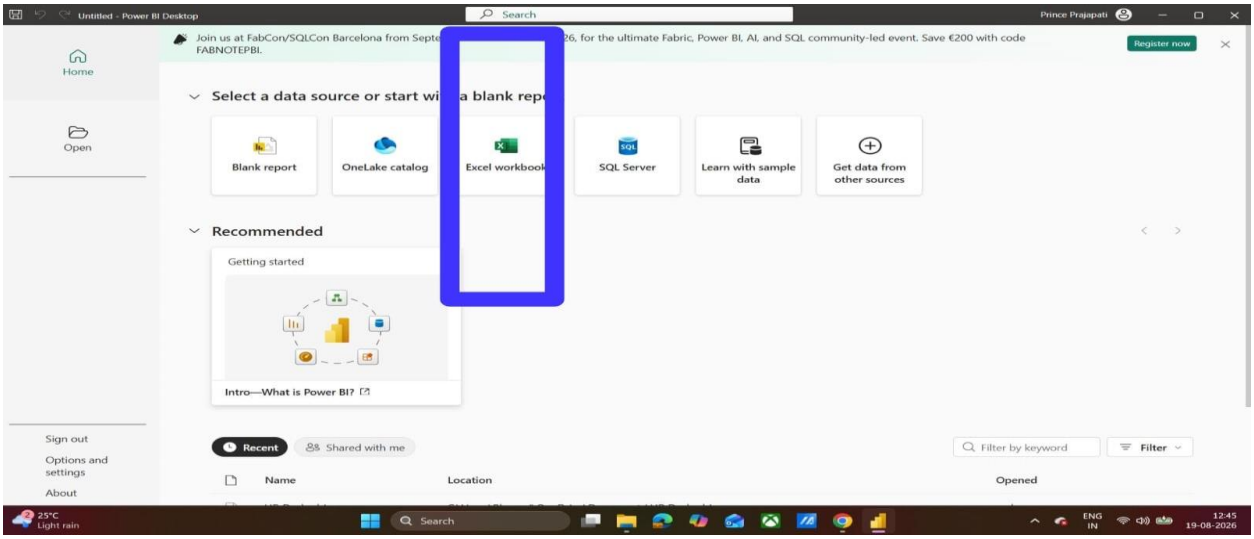
- Alternatively, if you are already on the main report view, click the Get Data icon located under the Home ribbon tab at the top toolbar.
- This action will launch the main Get Data window displaying all available data connector categories.

Get Data Option



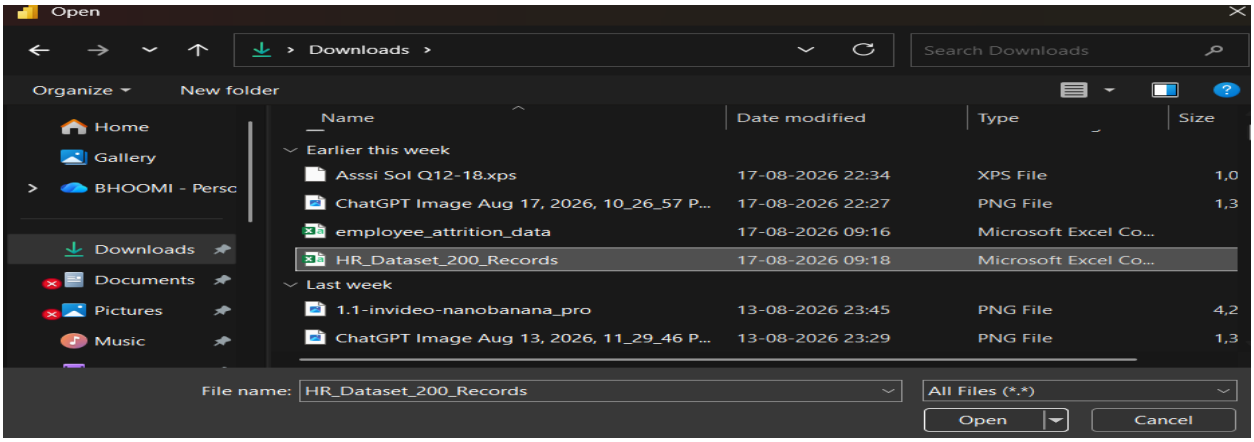
- The Get Data option opens a menu showing all available data connectors categorized by type, such as files and databases. It allows you to select your target format—like Excel or Text/CSV—and connect it directly to Power BI.

Step 4: Select Excel/CSV



- Inside the open Get Data window, click File from the left-hand category list, or stay on the All tab to view standard file options.
- Scroll through the available file connectors in the center pane and click on Excel Workbook (if importing a `.xlsx` file) or Text/CSV (if importing a `.csv` file, such as your `HR_Dataset_200_Records.csv` file).
- Once your preferred format is highlighted, click the Connect button located at the bottom right corner of the window to open your file browser.

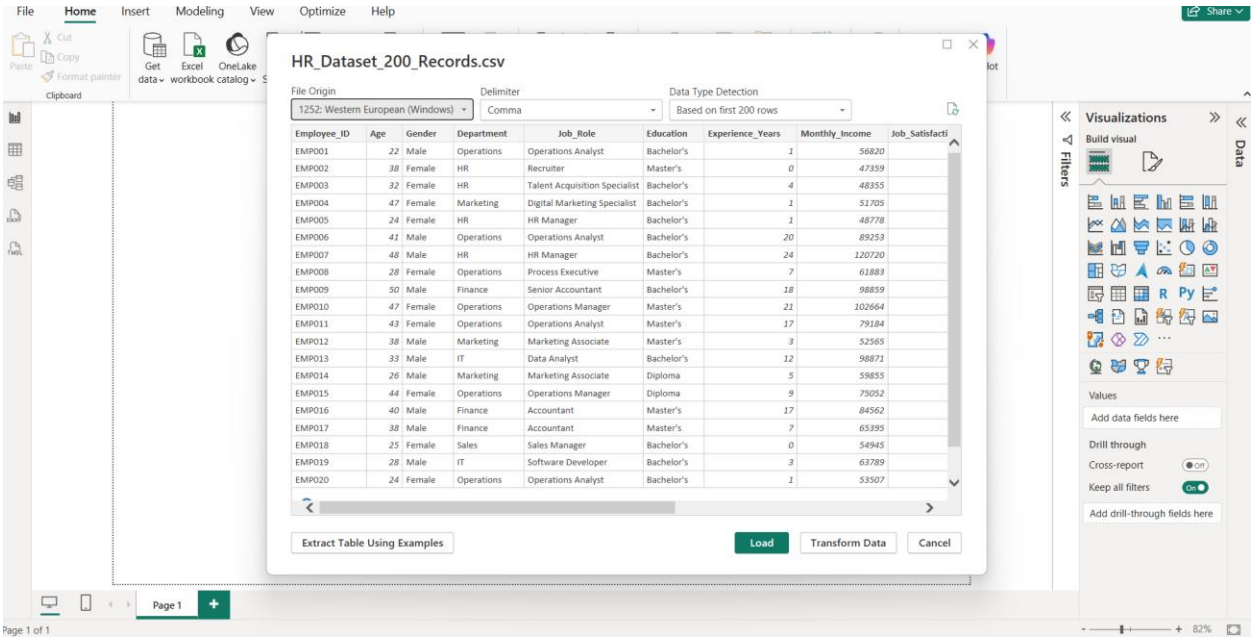
Step 5: Select Dataset



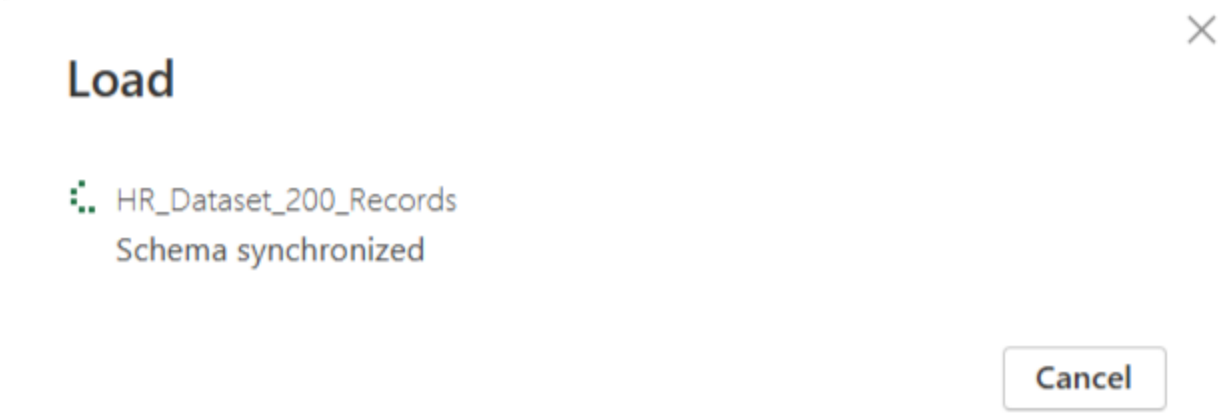
- After clicking Connect, a standard Windows Open file browser window will pop up on your screen.

- Navigate through your local directories and folders (for example, open the Downloads folder) to locate where your dataset is saved.
- Click on your targeted dataset file—such as `HR_Dataset_200_Records.csv` (or your specific Excel file)—to select it.
- Click the Open button at the bottom right corner of the file browser to initiate the data preview process in Power BI.

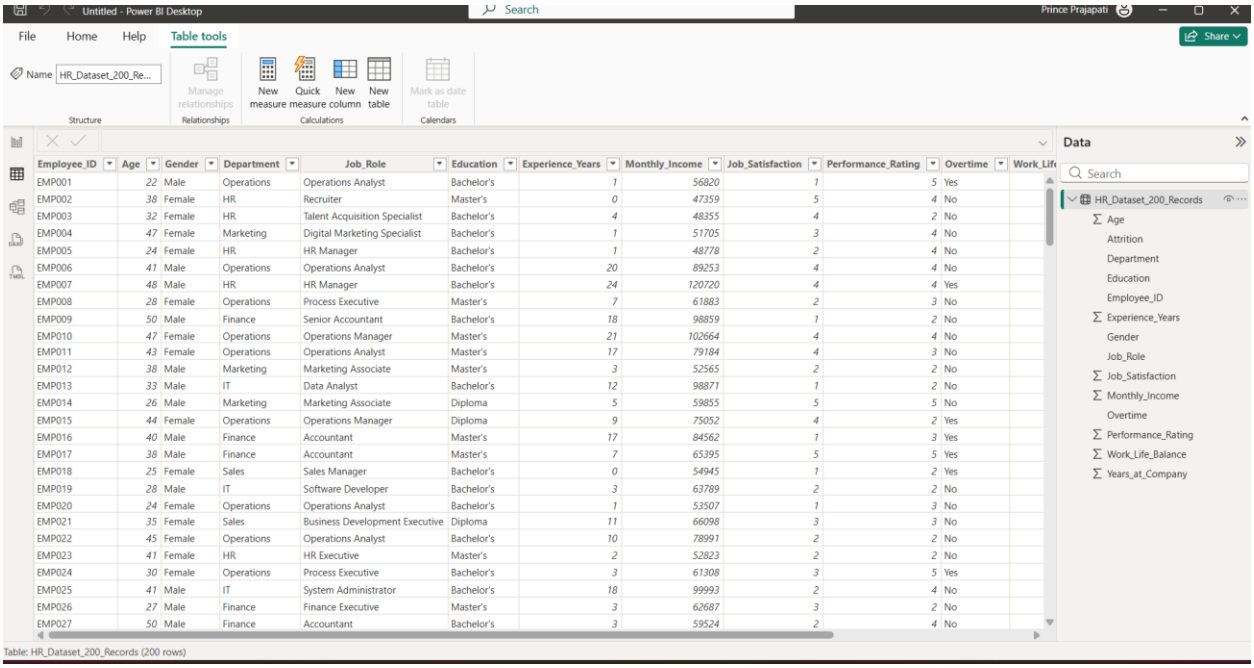
Step 6: Load Data



- After selecting your file, a preview window opens showing a sample of the data along with details like encoding, delimiter, and data type detection.
- Review the dataset to confirm that the columns and headers are aligned correctly.
- Click the green Load button at the bottom of the dialog box to import the data directly into your Power BI data model.

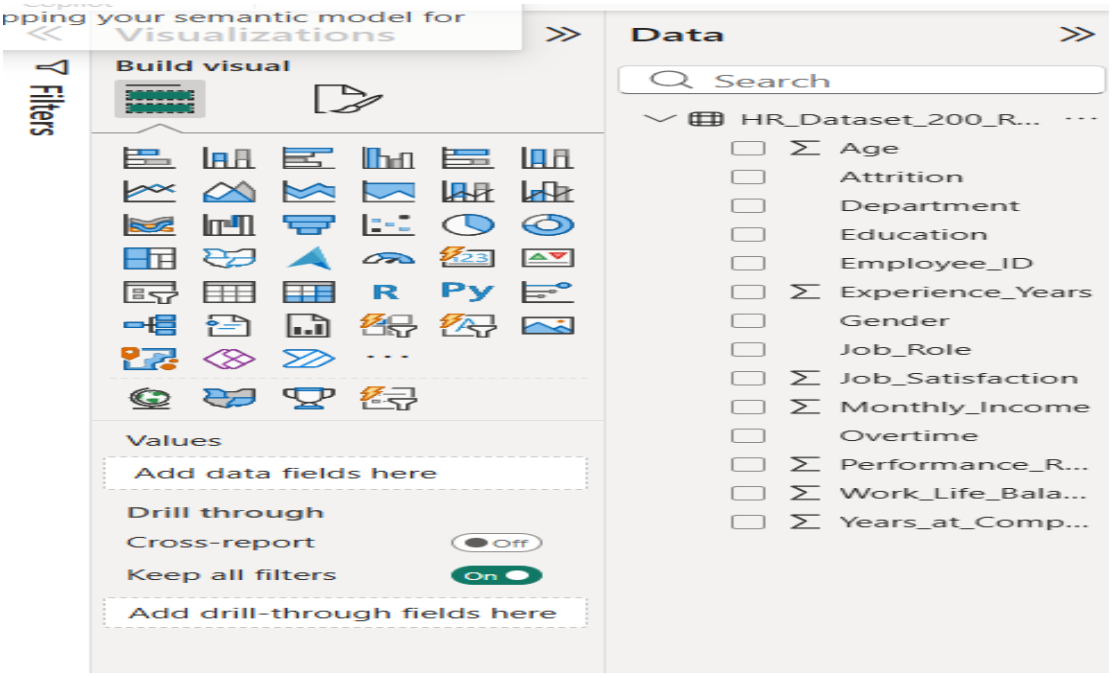


Step 7: Verify Imported Data



Employee_ID	Age	Gender	Department	Job_Role	Education	Experience_Years	Monthly_Income	Job_Satisfaction	Performance_Rating	Overtime	Work_Life
EMP001	22	Male	Operations	Operations Analyst	Bachelor's	1	56820	1	5	Yes	
EMP002	38	Female	HR	Recruiter	Master's	0	47359	5	4	No	
EMP003	32	Female	HR	Talent Acquisition Specialist	Bachelor's	4	48355	4	2	No	
EMP004	47	Female	Marketing	Digital Marketing Specialist	Bachelor's	1	51705	3	4	No	
EMP005	24	Female	HR	HR Manager	Bachelor's	1	48778	2	4	No	
EMP006	41	Male	Operations	Operations Analyst	Bachelor's	20	89253	4	4	No	
EMP007	48	Male	HR	HR Manager	Bachelor's	24	120720	4	4	Yes	
EMP008	28	Female	Operations	Process Executive	Master's	7	61883	2	3	No	
EMP009	50	Male	Finance	Senior Accountant	Bachelor's	18	98859	1	2	No	
EMP010	47	Female	Operations	Operations Manager	Master's	21	102664	4	4	No	
EMP011	43	Female	Operations	Operations Analyst	Master's	17	79184	4	3	No	
EMP012	38	Male	Marketing	Marketing Associate	Master's	3	52565	2	2	No	
EMP013	33	Male	IT	Data Analyst	Bachelor's	12	98871	1	2	No	
EMP014	26	Male	Marketing	Marketing Associate	Diploma	5	59855	5	5	No	
EMP015	44	Female	Operations	Operations Manager	Diploma	9	75052	4	2	Yes	
EMP016	40	Male	Finance	Accountant	Master's	17	84562	1	3	Yes	
EMP017	38	Male	Finance	Accountant	Master's	7	65395	5	5	Yes	
EMP018	25	Female	Sales	Sales Manager	Bachelor's	0	54945	1	2	Yes	
EMP019	28	Male	IT	Software Developer	Bachelor's	3	63789	2	2	No	
EMP020	24	Female	Operations	Operations Analyst	Bachelor's	1	53507	1	3	No	
EMP021	35	Female	Sales	Business Development Executive	Diploma	11	66098	3	3	No	
EMP022	45	Female	Operations	Operations Analyst	Bachelor's	10	78991	2	2	No	
EMP023	41	Female	HR	HR Executive	Master's	2	52823	2	2	No	
EMP024	30	Female	Operations	Process Executive	Bachelor's	3	61308	3	5	Yes	
EMP025	41	Male	IT	System Administrator	Bachelor's	18	99993	2	4	No	
EMP026	27	Male	Finance	Finance Executive	Master's	3	62687	3	2	No	
EMP027	50	Male	Finance	Accountant	Bachelor's	3	59524	2	4	No	

- Switch to the Data View tab on the left sidebar of Power BI Desktop to inspect the dataset in a complete grid format.
- Confirm that all rows and columns (such as `Employee_ID`, `Age`, `Department`, `Monthly_Income`, and `Attrition`) have loaded correctly without missing data.
- Check the Data pane on the right side to verify that field names are listed along with their appropriate data type symbols (e.g., the $\sum$ icon next to numerical fields).



8. Result

- Power BI Desktop was successfully installed, and data from the CSV file (`HR_Dataset_200_Records.csv`) was successfully imported and verified within Power BI.